Animals Use For Research Purpose

Objective:

Animals used in research by scientist and zoologist for various objectives:-

function

→To understand Physiology and pathology of man.

To diagnose the effects of various drugs, surgery procedures and treatment procedures on man.

Because the biology use Some animals which Include Mammals with humans. So we / Researchers use that mammals to study the biology of human being.

Mammals which mostly used for Research purpose Include:-

(1) Rabbit

(2) Frog (Amphibian)

(3) Rat

(4) Archaeopteryx

Detail of such animals for Research purpose are given one by one.

(1) Research on Frog:

Introduction:

(1) Frog belongs to Amphibians Groups .
(2) They evolved ~280 million years ago.(

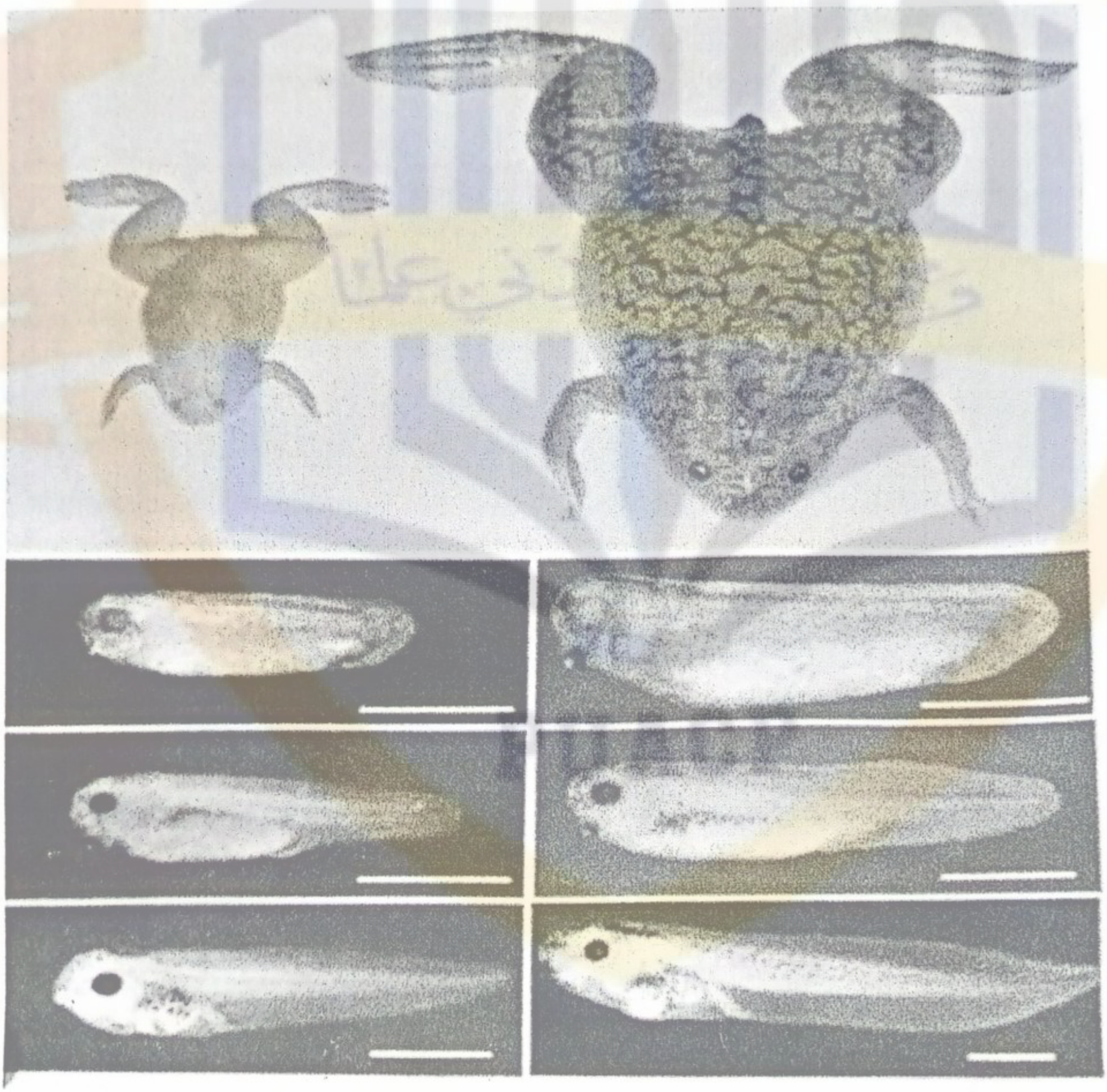
Frogs are cold blooded, vertebrate animal.

Features of Frog:

(1) Frogs have protruding eyes.
(2) External nares are present and Tympanum are also present.
(3) Frog have fore limbs, hind limbs and 5 Toes with four digits.
(4) Skin is soft and moist.

Research on Frogs: specie Include Xenopus laevis and Xenopus tropicalis.

Frogs which use for research purpose for various reasons and various fields of biology including:



x. tropicalis *X. laevis*

Developmental Biology:

- Frogs are used in developmental biology to understand the developmental stages and process of development.
- Frogs used in developmental biology because Frog embryo are externally developed and rapid produced.
- Scientific study the process of fertilization, embryonic development and organogenesis using frog as model.

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- Frogs eggs are Transparent which provide ease for the observation of embryonic development.

(1)Neurobiology:

Frogs large and accessible embryo facilitate the researcher's on observation of process Include:

- Neural crest migration
- Axon guidance
- Synapse formation.

All these processes helps in the observation of neurobiology of human.

(3)Toxicology:

- Frogs are used in toxicology studies to assess the effects of various chemicals, pollutants drugs on physiology and behavior or development of frogs.
- By observing the effectiveness of these toxin on Frogs. It will be easy to understand the efficacy of such toxin on humans.

(4)Regeneration:

- Some Frog species such as Xenopus laevis possess remarkable regenerative abilities. Scientist study these frogs to understand the regeneration. Mechanism and analyzed that which factors and gene play role in regeneration process.

(5)Ecology:

- Frogs are important indicators of environmental health and research on frog populations can provide insight to ecosystem dynamics. Habitat quality and impacts of environmental processes such as
- Pollution
- Habitat loss

- climate change

(6)Genetics:

- Frogs species such as Xenopus laevis and Xenopus tropicalis used as model for genetic study and research such as.
- Gene function
- Gene evolution
- gene regulatory networks. → regulatory interaction btw genes

Products obtained from Frogs:-

Major products obtained from frogs are toxin such as:

1. Alkaloids
2. Bufotoxin → Inflammation or irritation
3. Dermatoxins

4.Tetrodotoxins

5.Adenoregulins

(2) Research on Rabbits:

- Some areas of research involving rabbits are:

(1)Biomedical Research:

- Rabbits are commonly used in biomedical research to study various diseases, disorders, medical treatments.
- Rabbit serve as valuable models for studying cardiovascular diseases, respiratory, cancer, and neurological diseases.
  Researchers used Rabbits to investigate:
- Disease mechanism
- Test Therapeutic Interventions.
- Evaluate the safety and efficacy of drugs and medical devices.

(2)Pharmacology:

- Rabbits used as model to study the safety and efficacy of certain drugs, chemicals and environmental pollutants.
- Researchers used rabbits to evaluate drug metabolism, Pharmacokinetics, and toxological effects include acetic toxicity, chronic Toxicity and Reproductive toxicity.

(3)Immunology:

- Rabbits are widely used in immunological research to study Immune responses. Antibody production and vaccine development .
- They are commonly used to produce polyclonal antibodies for research and diagnostic and to evaluate the Immunogenicity and efficacy of various vaccines against infectious diseases such as Influenza, Hepatitis and human Papillomavirus (HPV).

liver

(4)Opthamdogy:

- Rabbits are commonly used in opthalmic research to study eye diseases, vision disorders and ocular drug delivery. vision loss blindness
- Researchers use rabbits to model conditions such as glaucoma, retinal degeneration cataracts and corneal injuries.
- Rabbits are used to evaluate the efficacy and safely of certain opthalmic drugs and treatments.

(7)Surgical Models:

- Rabbits are used as surgical models to develop and refine surgical models, techniques, procedures and medical devices.
- Rabbits used as model to stimulate surgical conditions practice surgical skills and evaluate the outcomes of surgical process in areas such as orthopedics, cardiology, and neurology.

(2) Research on Rats:

- Mice and rats served as a preferred species for biomedical research animal models due to their anatomical, physiological and generic similarity to humans.

Advantage of using rats as biomedical research:-

Advantage of using rats as research includes:

- Rats Small size
- ease of maintenance
- Short life cycle .
- Abundant genetic resources.

Since, mice or rat are small in size and generally cost less to maintains.

However, many areas of investigation where rats are preferred including:

(1) Cardiovascular disease / Research
(2) Behavioral studies.
(3) Toxicology.

(1)Cardiovascular disease:-

- Cardiovascular disease is the leading cause of death and morbidity in developing countries.
- Cardiovascular disease is multifactorial caused by combination of environmental and genetic factors.
- Many unique strains of rats have been generated that model the complex nature of human obesity, Diabetes and cardiovascular disease.

(2)Behavioral Study:

- Rats are commonly used for behavioral studies because their behavior better mimics the behavior seen in humans. For example:
    - Expansion of a three base pair sequence in the FMR1 gene is responsible for Fragile X syndrome. The most common cause of inherited intellectual disability in humans, is the expansion of FMR1 gene due to adding more base pair as a result FMR1 gene becomes methylated and knock out or not expressed.
    - This repression of gene cause Fragile X syndrome in rats and same in humans which ultimately slows down the behavioral activity of rats especially vocalizations during breading period.

(3) Research on Archaeopteryx:

Several fossils like Archaeopteryx found in Southern Germany about 150 years ago.

- They named as Archaeopteryx.
- These fossils possessed characteristics of both reptiles and birds.

Objective of Research On Archaeopteryx:-

Research on Archaeopteryx is significant for several reasons:

(1) Transitional Fossil:

- Archaeopteryx is considered as a Transitional fossil between non- avian dinosaurs and modern birds.
- Its combination of dinosaur like and modern bird like features provides crucial evidence supporting the theory of avian evolution from theropod dinosaurs.
- Research on Archaeopteryx help the scientist to understand sequence of evolutionary changes that occurred during the transition from dinosaurs to birds.

2-Origin of Flight:

- Archaeopteryxes is important for studying origin of flight in birds.

Fossils specimen of Archaeopteryx show evidence of feather. By studying the anatomy wing morphology and flight capabilities of Archaeopteryx to researchers gain insight to early stages of flight evolution and adaptation that enable the birds to reach and fly in sky.

3-Paleoecology:

Studies of Archaeopteryx provides insight to the ancient ecosystems, paleo environments and biodiversity during the late Jurassic period.

- Fossilized remains of Archaeopteryx and associated organisms offer clues about habitat preferences, diet, behavior and ecological Interactions in prehistoric ecosystem.

(4)Evolutionary Biology:

Research on Archaeopteryx contributes to air understanding of evolutionary processes include speciation, adaptation and morphological diversification.

- By comparing Archaeopteryx with other dinosaurs and early birds, scientist can infer evolutionary relationships, reconstructive ancestral traits and evolutionary change over time.

Characteristics of Archaeopteryx:

1- Wings/Feathers:

Archaeopteryx feathers are long and body and tail are also long. Despite its bird like feathers Archaeopteryx 0 retained contains theropod dinosaurs. These features include a long bony tail, clawed fingers on its wings, teeth in its jaws and reptilian lice pelvis.

2- Size:

Archaeopteryx is crow sized0.5 to 0.6m long and 1.6 to 2 feet in length. Its size is intermediate between those of Small theropod and modern bird.

3- Skull:

Archaeopteryx had a small bird like skull with pointed scout and large eye sockets. It had beak like structure formed by the fusion of its jaw bones.

4- Flight capability:

Its skeletal structure and muscles suggest that it had been limited to gliding or short bursts of powered flight rater than sustained light.